

A Crowdsourcing Platform for Constructing Accessibility Maps Supporting Multiple Participation Modes

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ABSTRACT

Accessibility maps enable impaired/elderly people to move around more smoothly and with less risk. However, very few examples satisfy both the accuracy and coverage requirements, owing to the high cost of physically auditing everchanging roads and pathways. Although crowdsourcing approaches can ostensibly solve this problem, existing studies have relied on volunteers with free time and high motivation. In this paper, we propose a crowdsourcing platform for constructing accessibility maps to support four participation modes: **Reporter** for people having plenty of free time and high motivation; **Gaming reporter** for people having plenty of free time but with low motivation; **Walker** for people lacking enough free time but with high motivation; and **Gaming walker** for people lacking enough free time and with low motivation. This design allows people to select a suitable participation method, depending on their time and motivation. In this study, we developed a prototype by integrating deep learning techniques, a game design theory, and heatmap visualization.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility systems and tools**; • **Information systems** → *Collaborative and social computing systems and tools*.

KEYWORDS

accessibility map, crowdsourcing, deep learning, gamification

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1 INTRODUCTION

There are many accessibility problems (e.g., stairs and slopes) that can hinder the free movement of impaired/elderly people (Figure 1). Identifying the types of such problems and their locations is beneficial for enabling the smooth movement of these people and urging governments to address these problems. Maps that provide accessibility information about roads and pathways (i.e., *accessibility maps*) can be used to these ends. However, very few examples satisfy both the accuracy and coverage requirements, owing to the high costs of conducting physical audits of everchanging roads and pathways.



Figure 1: Accessibility problems: (a) stairs without a ramp; (b) steep and obstructed pathway.

Some studies have attempted to detect accessibility problems manually by using road photos stored in street view services [3, 10, 12, 24]; others have used (semi-) automatic approaches [1, 13, 32]. However, vision-based approaches that rely on street view services are limited by inaccurate detection of accessibility problems as only one middle-resolution image is available for each point. Additionally, many objects often obscure the pathways because of the position of the photographers capturing the images.

For these reasons, we focus on a crowdsourcing approach [2, 4, 6, 9, 15, 16, 19, 22, 23, 28–30] that leverages people to acquire data in the field. Although this approach has been attempted previously, existing studies did not consider the various conditions

of participants; who may have varying degrees of free time and motivation. This has limited the number of data-collecting participants.

In this paper, to construct an accessibility map with higher accuracy and coverage, we propose a crowdsourcing platform called *BScanner*¹ that supports four participation modes (Figure2): (1) **Reporter** for people having plenty of free time and high motivation; (2) **Gaming reporter** for people having plenty of free time but with low motivation; (3) **Walker** for people lacking enough free time but with high motivation; and (4) **Gaming walker** for people lacking enough free time and with low motivation. This design allows people to select a suitable participation method, depending on their free time and motivation.

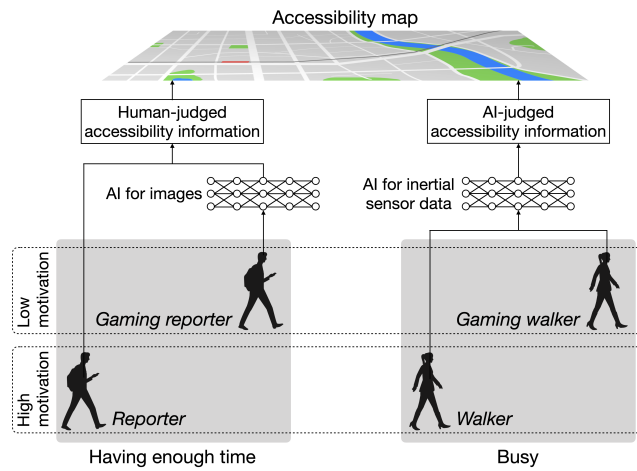


Figure 2: Proposed platform *BScanner*.

2 RELATED WORK

Several systems for constructing accessibility maps have been proposed in response to the growing awareness of accessibility problems. Such systems typically utilized road photos or situated crowdsourcing.

2.1 Road photos

By combining road photos stored in street view services and human resources, many researchers have tried to gather information about accessibility problems. Badland et al. [3] and Rundle et al. [24] measured the efficiency and reliability of virtual streetscape audits using a street view service. CrossingGuard [10] used OpenStreetMap and Google Street View (GSV) to gather detailed information on an intersection from online workers. Similarly, Hara et al. [12] used GSV to gather information about accessibility problems from online workers. Their system has evolved and is now provided as a web-based crowdsourcing tool [26].

By applying computer vision techniques to road photos stored in street view services, other researchers have attempted to automatically detect accessibility problems. Zebra crossing spotter [1]

detected and localized zebra crosswalks, by analyzing street view panoramas and satellite images obtained from Google services. Sun et al. [32] proposed a model based on a convolutional neural network structure to judge whether curb ramps were missing at an intersection. Tohme [13] adopted a semi-automatic approach to detect curb ramps in a street view service using both computer vision techniques and human workers.

These vision-based works, which relied on street view services, were limited by the inaccurate detection of accessibility problems for a variety of reasons. First, only one middle-resolution image (640 × 640 pixels for GSV) captured from a car is available for each point. Second, many objects (e.g., people, cars, and trees) obscure the pathway surfaces. Finally, images are often outdated owing to rapid developments in big cities and the low frequency of image capturing in rural areas. These factors limit the accuracy and coverage of the manual/automatic detection of accessibility problems.

2.2 Situated crowdsourcing

2.2.1 Manual reporting approach. Historically, professional investigators have audited pathways in the field. Many metrics for assessing accessibility [5, 31] have been introduced. These traditional approaches have been highly accurate, but they are impractical to apply to large areas, owing to the significant costs involved. However, with the widespread use of smartphones, manual reporting approaches by volunteers have been proposed. Approaches such as SeeClickFix [28] and IBM Sidewalks [4, 29] allowed people to report sidewalk problems, including missing curb ramps. Wheelmap [30] and WheelLog! [19] are web services that allow people to upload geotagged accessibility information for wheelchair users. Goncalves et al. [9] evaluated a similar system, showing that visual and location cues had an effect on participants' attitudes and awareness regarding disability and accessibility problems.

In terms of accuracy, these human assessment approaches are preferable. Citizen reporting approaches can also reduce cost burdens. However, these approaches rely on people who have sufficient free time and are highly motivated. This restricts the coverage of accessibility information.

2.2.2 Sensor-based approach. Recently, sensor-based approaches for constructing accessibility maps have attracted attention. Prandi et al. [23] and Aly et al. [2] presented a scheme for gathering accessibility information by analyzing sensor data obtained from mobile devices (e.g., smartphones). Miyata et al. [16] proposed an approach for detecting accessibility problems that used a neural network model trained using acceleration data acquired from the smartphones of able-bodied pedestrians. Kurauchi et al. [15] introduced a two-step machine learning model; it first recognized the mode of transportation (e.g., walking, wheelchair, or stroller), and then recognized the type of problem by analyzing sensor data.

In terms of coverage, these sensor-based approaches are preferable. However, they are still designed based on the assumption that people are highly motivated to install a sensing application on their smartphones and run it while walking.

2.2.3 Gamification approach. Gamification has been defined as the use of game design elements in non-game contexts [8], and a

¹This name represents "barrier scanner," and the participants collect information similar to "bees" collecting nectar.

process of enhancing a service with affordances for gameful experiences in order to support user's overall value creation [14]. To increase user motivation and engagement for activities, gamification has been used in various contexts including education, work, ideation, social networks, orientation, and data gathering [11, 27]. In the area of situated crowdsourcing, some gamification systems, such as CityConqueror [21] and City Explorer [20], have been proposed. From the perspective of allowing users to report accessibility problems, Kijk! [6] has provided badges as rewards for reporting problems, whereas Geo-Zombie [22] has required users to report accessibility problems to obtain weapons and ammunition to shoot virtual zombies.

In terms of motivation, these gamification approaches are preferable. However, these approaches require a great deal of manual user configuration (e.g., taking photos of problems and describing the problems). This can limit the number of participants. Furthermore, because game users are often less motivated to provide intrinsic results (i.e., reports of accessibility problems), a mechanism that can be used to evaluate the quality of their data is required.

3 SYSTEM REQUIREMENTS AND DESIGN

3.1 System requirements

Our study aims to construct an accessibility map that satisfies both the accuracy and coverage required. As described, current accessibility maps do not satisfy both of these needs. Crowdsourcing approaches could solve this problem; however, existing studies have been designed for specific types of participants (e.g., highly motivated people who have free time). To achieve our goal, we need to develop a system that meets the following requirements.

R1. Manual reporting. Presently, there is no automated technology that outperforms manual work regarding the accuracy of finding accessibility problems. Thus, it is important to provide a method for participants to manually report such problems.

R2. Hassle-free. Busy people, even if motivated, are less likely to participate in time-consuming activities. Thus, it is important to provide a hassle-free method for participants to contribute to detecting accessibility problems.

R3. Entertainment. People having low motivation, even if they have time, are less likely to participate in activities that are not related to their own pursuits. Thus, it is more effective to provide an entertaining method to capture their attention.

R4. Validation. This requirement relates to R3. People having low motivation can often provide low quality data. A validation mechanism is required to deal with such low quality data.

3.2 System design

To meet these requirements, we design a platform called *BScanner* [18], shown in Figure 2. This platform provides a smartphone application with four modes, namely, *Reporter*, *Gaming reporter*, *Walker* and *Gaming walker*, as shown in Figure 3. There are two main data paths. One applies to human-judged data; this path is suitable for participants having plenty of time as manual reporting can be time-consuming. The other path applies to AI-judged data; relatively busy participants will prefer this path as detailed manual reporting is not required. Each data path is equipped with a gamification mode to encourage the less motivated participants; each

game is designed around the theory of computer games [7] and the findings of location-based games [6, 20–22]. This design allows participants to select a suitable participation method, depending on their free time and motivation. To the best of our knowledge, the proposed platform is the first to address all of these requirements together, as described in Table 1.

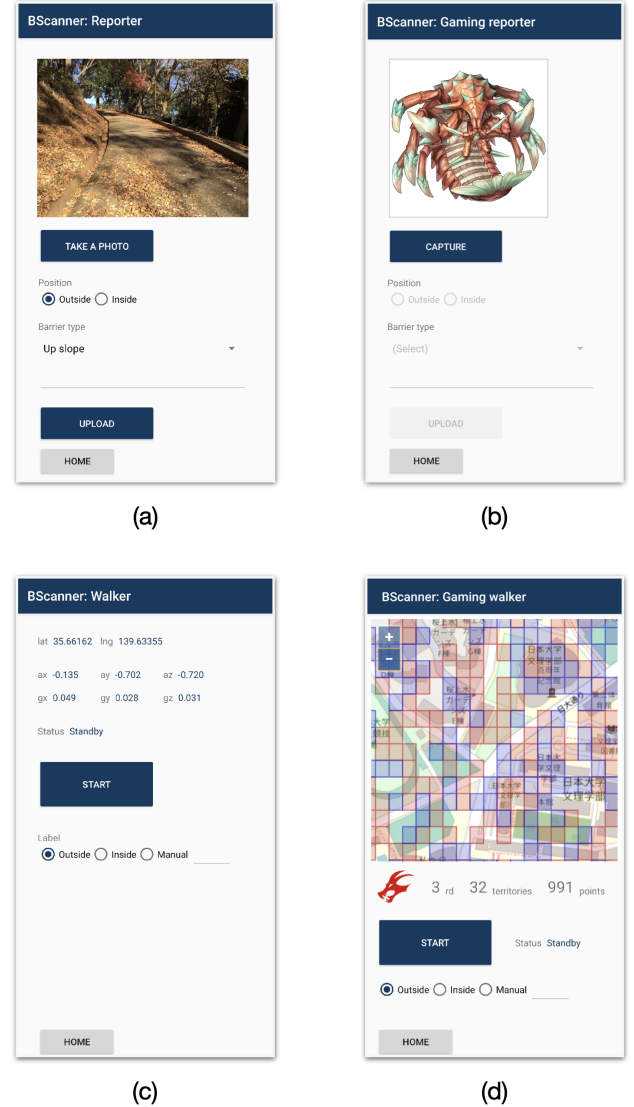


Figure 3: (a)*Reporter*, (b)*Gaming reporter*, (c)*Walker* and (d)*Gaming walker*.

3.2.1 Reporter. This mode (Figure 3 (a)) applies to participants having plenty of free time and high motivation. Accessibility problems are manually reported by participants using photos, descriptions, and location information (R1).

Participants employing this mode act as follows. First, they audit roads and pathways and find an accessibility problem. Second, they

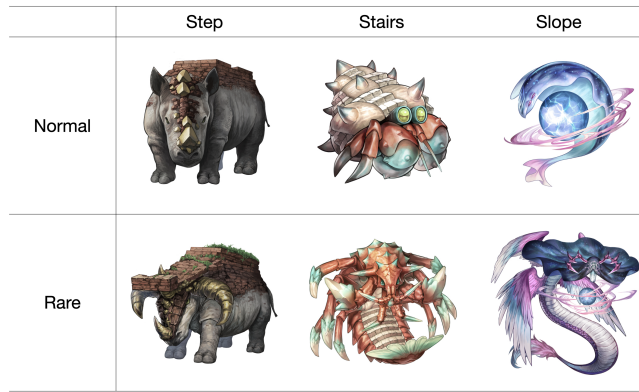
Table 1: Comparison of situated crowdsourcing systems for constructing accessibility maps.

Systems/works	Requirements			
	R1. Manual reporting	R2. Hassle-free	R3. Entertainment	R4. Validation
SeeClickFix [28], IBM Sidewalks [4, 29], Wheelmap [30], WheelLog! [19] and Goncalves et al. [9]	✓	-	-	-
Aly et al. [2], Miyata et al. [16] and Kurauchi et al. [15]	-	✓	-	-
Prandi et al. [23]	✓	✓	-	-
Kijk! [6] and Geo-Zombie [22]	✓	-	✓	-
BScanner	✓	✓	✓	✓

take a photo of the problem. Third, they select an appropriate label (e.g., stairs or slope) for the problem from a list of options. Finally, they upload the photo and the label with location information measured by their smartphone's global positioning system (GPS) unit.

3.2.2 Gaming reporter. This mode (Figure 3 (b)) provides the gamification mode of *Reporter* for participants having plenty of free time but with less motivation. Accessibility problems are manually reported by participants via photos and location information (R1). In this game, participants collect *monsters* by taking photos of accessibility problems (R3). Each monster corresponds to an accessibility problem, as shown in Figure 4. An image analyzer (*AI for images* in Figure 2) estimates the type of accessibility problems contained in the photo. If no type of problem is detected, the participant does not secure a monster (R4).

Participants employing this mode act as follows. First, they audit roads and pathways and find an accessibility problem. Second, they take a photo of the problem. Third, they upload the photo with location information measured by their smartphone's GPS unit. Finally, they obtain a monster corresponding to the problem.

**Figure 4: Monsters appearing in *Gaming reporter*. Each shape represents each accessibility problem.**

3.2.3 Walker. This mode (Figure 3 (c)) applies to participants lacking enough free time but having high motivation. Accessibility problems are detected by analyzing sensor data obtained from walkers' smartphones. Participants are not required to find or describe accessibility problems manually (R2).

Participants employing this mode act as follows. First, they launch the application in this mode. Second, they place the smartphone in their trouser pocket. Finally, they walk for their own purposes (e.g., commuting or shopping). At this time, the application records their acceleration, angular velocity, and location information (latitude and longitude) at a frequency of 20 Hz; these variables are measured by the smartphone's accelerometer, gyroscope, and GPS unit, respectively.

Our system analyzes the obtained data as follows. First, the system divides all data into small pieces of data; the length of each piece is equivalent to three seconds of walking. Second, the system analyzes each piece of data and estimates the type of road, using a neural network model for detecting accessibility problems [17] (Figure 5 (left)). Currently, this model can estimate seven road types (i.e., *flat*, *up step*, *down step*, *up stairs*, *down stairs*, *up high slope*, and *down high slope*) with an F-measure of 0.89, as shown in Figure 5 (right). Note that this performance may vary, depending on the position and orientation of the smartphone. Third, the system associates each estimated road type with the location information. Finally, the system calculates the probability of the existence of accessibility problems for each point. Specifically, it calculates the ratio of data estimated as problems (i.e., *up/down step*, *up/down stairs*, or *up/down high slope*) to all of the data associated with the point.

3.2.4 Gaming walker. Developed with reference to a similar system [33], this mode (Figure 3 (d)) is the gamification mode of *Walker* for participants lacking enough free time and with low motivation. Accessibility problems are detected by analyzing sensor data obtained from their smartphones in the same manner as that of *Walker*. Participants do not need to make accessibility reports manually (R2); they only need to play a location-based game (R3) that compels them to walk throughout the real environment. The goal of the game for the player is to expand their territories; players are divided into two teams (blue vs red) to compete in obtaining a larger area of territory by walking. If they do something different, such as running (which reduces the estimation performance of the neural network), instead of walking, they cannot expand their territories (R4).

Participants employing this mode act as follows. First, they find a virtual area to be obtained using the game map that shows the territory of the blue and red teams. Second, they go to the real area. Third, they place their smartphone in their trouser pocket. Finally, they walk to obtain the virtual area. At this time, the application records the data in the same format as that recorded by *Walker*.

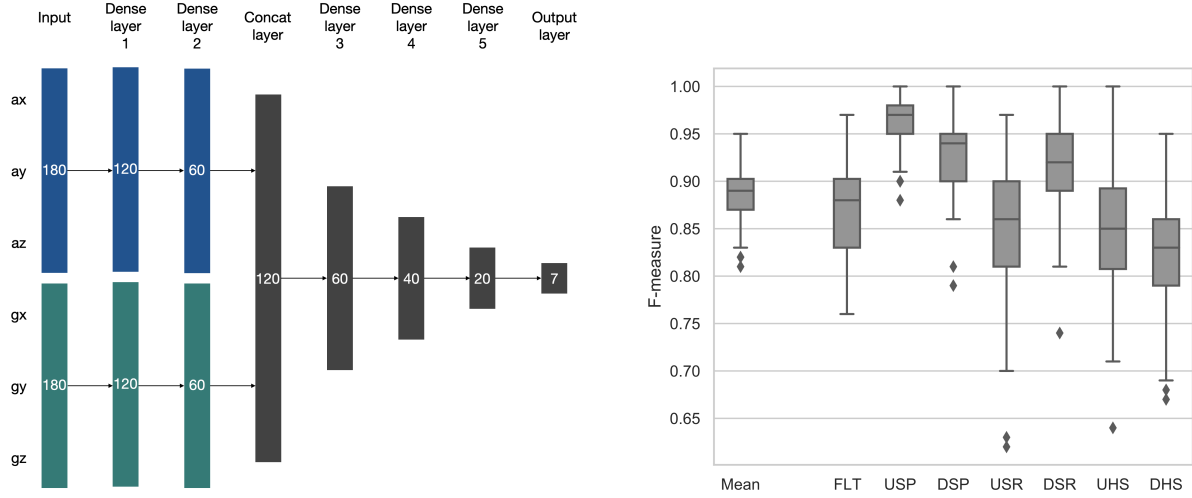


Figure 5: Left: the neural network model used for detection of accessibility problems. The input comprises $a[x, y, z]$ and $g[x, y, z]$. ax , ay , and az denote the acceleration sensor data along the x, y, and z axes, respectively. gx , gy , and gz denote the gyroscope data along the x, y, and z axes, respectively. The output is an estimated type of accessibility problems. We use mainstream techniques for model construction: Adam for parameter optimization; Hyperopt for hyperparameter optimization; Batch normalization for normalization; and Early stopping and Reducing learning rate on plateau for controlling the training plan. Right: F-measures for each road type (100 samples obtained from 10 participants for each type). FLT: walking on a flat sidewalk paved with asphalt. USP/DSP: walking up/down a step of which height is about 15 cm. USR/DSR: walking up/down stairs of which each step's height is about 15 cm. UHS/DHS: walking up/down a high slope of which angle is about 10° .

3.2.5 Visualization. As mentioned, our platform collects two types of data: human-judged data from *Reporter/Gaming reporter* and AI-judged data from *Walker/Gaming walker*. Because each data differs in character, using the same visualization method for all of them is not appropriate. This leads us to employ a visualization method that suits the character of each data item in a hybrid way, as shown in Figure 6. Human-judged data are discrete and have a relatively small number; each data item is displayed on the map as an icon linked to a description or photo uploaded from *Reporter/Gaming reporter*. In contrast, AI-judged data generated by analyzing time series sensor data uploaded from *Walker/Gaming walker* are continuous and have a relatively large number; the data distribution is displayed in a heatmap format, where the color represents the probability of the existence of accessibility problems. The red-colored parts indicate areas having a high probability of the existence of accessibility problems, such as up/down steps, up/down stairs, and up/down high slopes.

4 CONCLUSION AND FUTURE WORK

This paper details our work-in-progress crowdsourcing platform for constructing accessibility maps to support four modes of participation. This design allows people to select a suitable participation method, depending on their time and motivation. We developed a prototype by integrating deep learning techniques, game design theory, and heatmap visualization.

The main limitations of our work are as follows. First, *Walker* and *Gaming walker* are designed for human walkers; who do not use aisles dedicated to wheelchair users. This design make it more

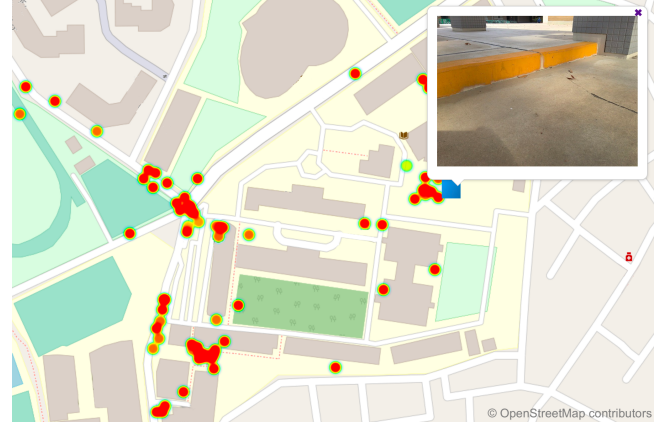


Figure 6: Example of our accessibility map.

difficult to gather sufficient data for such pathways. However, we can still gather a sufficient number of data from people using *Reporter*. Second, *Walker* and *Gaming walker* assume that participants place their smartphone in their trouser pocket. In real settings, the position of the smartphone will vary among individuals. To support various smartphone positions, neural network models for each position are required.

For future works, we plan to conduct two major studies. First, we plan to improve the quality of the games. To keep low motivated participants absorbed in the game, collaboration with professional

game creators will be indispensable. Second, we plan to conduct a case study in which we will allow public access to our prototype, to gather feedback from people in need of accessibility maps, including impaired/elderly. As reported by Project Sidewalk [25], accessibility problems have socio-political aspects. To make our platform truly practical, collaboration with government organizations will be required.

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